MIL-A-85097/10A(AS)
17 October 1985
SUPERSEDING
MIL-A-85097/10(AS)
26 June 1980

MILITARY SPECIFICATION SHEET

AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS); ROCKET MOTOR, UNDER SEAT, MECHANICALLY INITIATED, MARK 54 MOD 0 (LOADED ASSEMBLY)

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the Mark 54 Mod 0 Rocket Motor described herein shall consist of this specification and the latest issue of MIL-A-85097.

1. SCOPE

1.1 Scope. This specification sheet covers requirements unique to the Mk 54 Mod 0 Mechanically Initiated Under Seat Rocket Motor (loaded assembly) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents are in addition to those listed in the cover specification, MIL-A-85097, and form a part of this specification sheet to the extent specified herein.

DRAWINGS

Naval Air Systems Command (Code Ident 30003)

4903620	Label, Identification, Rocket Motor
2434829	Rocket Motor, Mark 54 Mod 0

3. REQUIREMENTS

3.1 General. The requirements in this specification sheet, unless in direct conflict with the basic specification, shall be considered additional, not substitute. The Rocket Motor shall conform to Drawing 2434829.

3.2 Preproduction.

3.2.1 Propellant aging. When tested in accordance with MIL-A-85097, the propellant shall meet the limit requirements made available by the contractor and accepted by the Government.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1377

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3.3 Preproduction/production.

3.3.1 Propellant. The propellant shall conform to the contractors' in-house propellant specification when tested for the data specified in MIL-A-85097. The contractor shall make available the results of the tested values and the in-house specification limits for comparison.

3.3.2 Life. Life shall conform to MIL-A-85097 with the exception of the propellant life which shall be not greater than 3 years on sheet stock and 1 year on extruded propellant grain.

3.3.2 Ballistic performance. The rocket motor shall meet the requirements of table I.

TABLE I. Ballistic performance requirements.

Parameter	−40° ± 5°F		160° ± 5°F	
	Min	Max	Min	Max
Rocket motor ignition delay (ms)	-	25	-	25
Maximum rocket motor thrust (lbf)	4250	6950	4250	6950
Rocket motor action time (ms)	220	520	220	520
Rocket motor impulse (lbf-s)	1275	1485	1275	1485

4. QUALITY ASSURANCE PROVISIONS

4.1 Examinations and tests.

4.1.1 Preproduction.

4.1.1.1 Environmental.

4.1.1.1.1 Vibration. For the AEPS covered by this specification sheet, the following test procedures shall be used in lieu of those specified in the general specification. Two rocket motors shall be subjected to vibration testing. The rocket motors shall be temperature conditioned for not less than 6 hours, one at −40° ± 5°F and the other at 160° ± 5°F. The rocket motors shall be maintained at specified conditioning temperature during these tests. If the rocket motor must be removed from the conditioning chamber or if the rocket motor temperature falls outside the specified temperature limits for more than 5 minutes, the testing shall stop and the rocket motor shall be reconditioned. A reconditioning time of 2 minutes for every minute the rocket motor spent outside the specified temperature envelope shall be required, up to a maximum of 8 hours. A continuous temperature history of the rocket motor shall be kept for all conditioning and test time. Excitation shall be applied through the aircraft seat attachment fittings unless otherwise noted in the contract. Accelerometers to measure acceleration along the axis perpendicular to the longitudinal axis of the rocket motor shall be placed midway between the attachment points. Longitudinal vibration shall be measured at either end of the rocket motor.

4.1.1.1.1.1 Resonance survey. Resonance surveys shall be conducted on both rocket motors in three axes at the maximum and minimum conditioning temperatures. Input vibratory acceleration of 5 g's shall be applied successively along each of the axes noted above when sweeping between 10 and 499 hertz. Input vibratory accelerations of 3 g's shall be applied for sweeps between 500 and 2,000 hertz. All resonances shall be as noted. During the resonance survey a permanent, continuous record of input vibratory acceleration and of unit response to the input vibratory acceleration shall be kept. This record shall be made on an instrument acceptable to the Government and shall include the response along each of the three axes. Frequency shall be varied at a rate not greater than 0.25 octave per minute.

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a. A resonance shall be considered as a frequency at which a clearly defined peak in the amplitude ratio (ratio of the output acceleration in any of the three igniter cartridge axes to the input acceleration as measured at the excitation attachment points) is greater than a value of 1.5 to 1.

b. If the resonance survey indicates a frequency band wherein the amplitude ratio is greater than 1.5 to 1 (as contrasted to a clearly defined peak) this entire band width shall be considered one resonant point. The minimum frequency band width which shall be considered a resonant band shall be 0.33 octave. If, also within this band width, there are clearly defined amplitude peaks, the most severe of these shall be considered an additional resonant point.

4.1.1.1.1.2 Endurance vibration. One rocket motor shall be conditioned to $-40° \pm 5°F$ and one to $160° \pm 5°F$. Each rocket motor shall be vibrated for 24 hours, 8 hours in each of the three axes. The vibration time in each axis shall be equally divided between the resonant frequencies noted in that axis during the resonant survey. A peak input acceleration of 5 g's shall be maintained at frequencies between 10 and 499 hertz and 3 g's for frequencies between 500 and 2,000 hertz. If no resonant frequency is detected in an axis, the rocket motor shall be vibrated in that axis for a minimum of 2 hours at 5 g's input through the 10- to 499-hertz range and 6 hours at 3 g's input through the 500- to 2,000-hertz range. Total vibration time for the rocket motors shall be 24 hours which shall include resonant and nonresonant frequencies, as applicable. At the conclusion of the tests, the rocket motors shall be visually and radiographically inspected for damage without disassembly of the rocket motor. The rocket motors shall be fired at the same temperatures they were conditioned and vibrated.

4.2 Preproduction/production.

4.2.1 Conditioning temperatures. The conditioning temperatures and tolerance shall be as follows:

a. Cold: $-40° \pm 5°F$
b. Hot: $160° \pm 5°F$

4.2.2 Conditioning period. Time for conditioning prior to static firing or during temperature cycling shall be as follows:

a. Static firing: not less than 6 hours
b. Temperature cycling: 6 to 8 hours

5. PACKAGING

(See latest revision of MIL-A-85097.)

6. NOTES

6.1 Intended use. The rocket motor is intended for use as a means of ejecting personnel from the RF/F-8 aircraft.

6.2 Definitions.

6.2.1 Rocket motor ignition delay. Rocket motor ignition delay is defined as the elapsed time from removal of the sear from the firing mechanism (as indicated by breakwire failure) to the first indication of rocket motor thrust.

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6.2.2 **Maximum rocket motor thrust.** The maximum rocket motor thrust is obtained by measuring the maximum vertical rocket thrust.

6.3 **Changes from previous issue.** Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - AS
(Project 1377-N770)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-85097/10A(AS)	(AEPS); Rocket Motor Under Seat,MARK54 MOD0 (Loaded Assembly)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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